ONE-PAGE BRIEF

# Post-quantum cryptography review: noir @ 5c98fecd3fe6

Client: Aztec / Noir — public repository (passive review) · Source: github.com/noir-lang/noir, commit `5c98fecd3fe6` · Reviewed: 2026-10-28 12:43 UTC · Risk profile: Web3 project

POST-QUANTUM MATURITY

**Level 0 of 4**

no post-quantum cryptography

FINDINGS

**0** Critical **2** High **5** Medium **3** Low **3** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Signatures: 10 + 5 − 12 = 3 → **too late**

**Next step:** Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

## WHAT MATTERS MOST

| | |
|---|---|
| High<br>QSI-2026-04754 | **quick-xml 0.26.0 is affected by RUSTSEC-2026-0194**<br>What to do: Upgrade to a fixed version (0.41.0). |
| High<br>QSI-2026-04755 | **quick-xml 0.26.0 is affected by RUSTSEC-2026-0195**<br>What to do: Upgrade to a fixed version (0.41.0). |
| Medium<br>QSI-2026-04747 | **Quantum-vulnerable Pedersen commitment: threshold signing and MPC protocols (2 file(s))** preliminary<br>What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys... |
| Medium<br>QSI-2026-04745 | **Quantum-vulnerable BLS12-381: role not determined (2 file(s))** preliminary<br>What to do: Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an abstraction layer for... |

6 more in the full report.

## WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

| Role | Algorithms | What to do |
|---|---|---|
| Medium threshold signing and MPC protocols | Pedersen commitment | Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the... |
| Medium role not determined | BLS12-381, ECDSA | Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an... |

## KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 7; unlikely to apply to this build: 1 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 6.

Static review of source code and manifests; files analysed: 783. No requests were sent to the owner's systems.

Full report, content SHA-256: `2512370e8b29a06abeaab2de23760dd081e1e98edb7e4afa39272d2dc1a07889`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.